TMDB - Box Office Prediction

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In [ ]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from sklearn.preprocessing import MultiLabelBinarizer, OneHotEncode
from xgboost import XGBRegressor
from sklearn.model_selection import train_test_split, RandomizedSea
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score
from wordcloud import WordCloud
import ast

%matplotlib inline
random_state = 42
test_size = 0.2
np.random.seed(random_state)
```

1. Data Loading and Overview

In [ ]:
```
train = pd.read_csv('./data/train.csv', index_col='id')
test = pd.read_csv('./data/test.csv', index_col='id')

print(f"Shape of the train set: {train.shape}")
print(f"Shape of the test set: {test.shape}")
```

```
Shape of the train set: (3000, 22)
Shape of the test set: (4398, 21)
```

In [ ]:
```
print(train.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3000 entries, 1 to 3000
Data columns (total 22 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   belongs_to_collection  604 non-null    str
 1   budget                 3000 non-null   int64
 2   genres                 2993 non-null   str
 3   homepage               946 non-null    str
 4   imdb_id                3000 non-null   str
 5   original_language      3000 non-null   str
 6   original_title         3000 non-null   str
 7   overview               2992 non-null   str
 8   popularity             3000 non-null   float64
 9   poster_path            2999 non-null   str
 10  production_companies   2844 non-null   str
 11  production_countries   2945 non-null   str
 12  release_date           3000 non-null   str
 13  runtime                2998 non-null   float64
 14  spoken_languages       2980 non-null   str
 15  status                 3000 non-null   str
 16  tagline                2403 non-null   str
 17  title                  3000 non-null   str
 18  Keywords               2724 non-null   str
 19  cast                   2987 non-null   str
 20  crew                   2984 non-null   str
 21  revenue                3000 non-null   int64
dtypes: float64(2), int64(2), str(18)
memory usage: 515.8 KB
None
```

As we can see, there are many columns with null values and some of them are of type object, which means they contain text data. We will need to handle these null values and possibly extract features from the text data later on.

In [ ]:
```
display(train.head())
```

id	belongs_to_collection	budget	genres	homepag
1	[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[{'id': 35, 'name': 'Comedy'}]	Nal
2	[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	Nal
3	NaN	3300000	[{'id': 18, 'name': 'Drama'}]	http://sonyclassics.com/whiplash
4	NaN	1200000	[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com
5	NaN	0	[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	Nal

5 rows × 22 columns

As we can see, there are some columns that contain lists of dictionaries, so we'll need to extract the needed information from those ones.

1.1 Dictionaries Extraction

In [ ]:
```
def parse_json(x):
    if pd.isna(x) or x == '':
        return []
    if isinstance(x, list):
        return x
    if isinstance(x, str):
        return ast.literal_eval(x)
    return []

def extract_from_json(df, column, key):
    return df[column].apply(
```

```
        lambda x: [d[key] for d in parse_json(x) if isinstance(d, d
    )

def extract_role_from_crew(df, role):
    temp = df.copy()
    return temp['crew'].apply(
        lambda x: next(
            (m['name'] for m in ast.literal_eval(x) if m.get('job')
            None
        ) if pd.notna(x) else None
    )

def extract_dictionaries(df):
    temp = df.copy()

    temp['belongs_to_collection'] = extract_from_json(temp, 'belong
    temp['genres'] = extract_from_json(temp, 'genres', 'name')
    temp['production_companies'] = extract_from_json(temp, 'product
    temp['production_countries'] = extract_from_json(temp, 'product
    temp['spoken_languages'] = extract_from_json(temp, 'spoken_lang
    temp['Keywords'] = extract_from_json(temp, 'Keywords', 'name')
    temp['cast'] = extract_from_json(temp, 'cast', 'name')

    # Crew column contains different roles, let's extract directors
    temp['director'] = extract_role_from_crew(temp, 'Director')
    temp['crew'] = extract_from_json(temp, 'crew', 'name')

    return temp

train = extract_dictionaries(train)
test = extract_dictionaries(test)
```

1.2 Release Dates Parsing

Firstly, we will analyze the distribution of the release date feature, which will be used also to handle the next features. We will extract the year, month, and day of the week from the release date and analyze their relationship with revenue.

As we can see the Release Dates are in the mm/dd/yy, but there are both films released before and after 2000 in the dataset. That said, we'll convert the dates into mm/dd/yyyy format, to handle better the release years. Note that the competition started in 2019, so we can consider that there are no movies released later than this year.

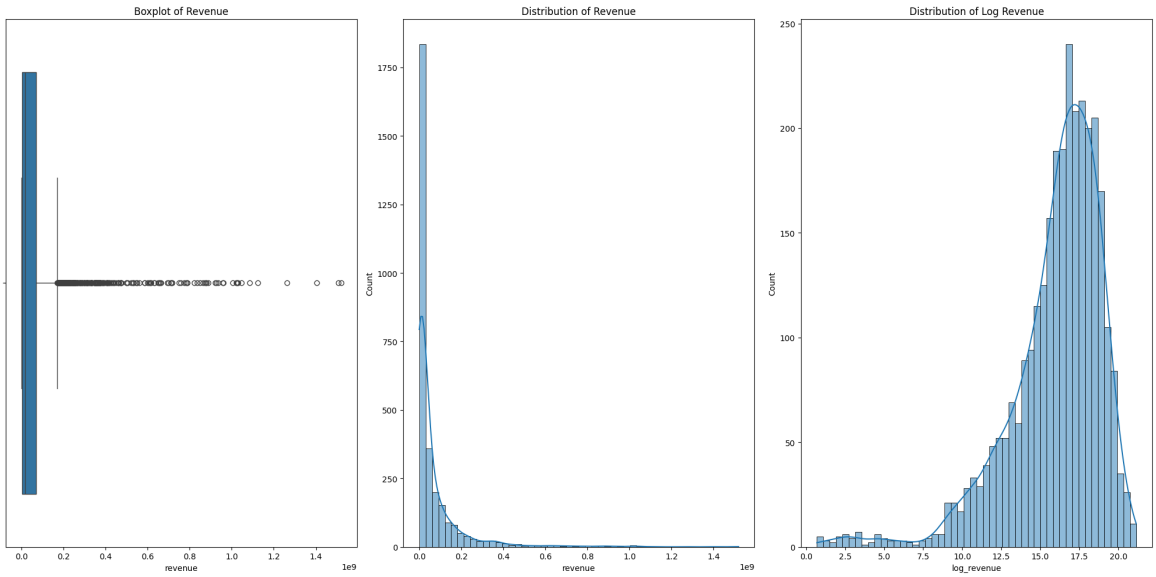
In [ ]:
```
def parse_release_dates(df):
    df = df.copy()
    df['release_date'] = pd.to_datetime(
        df['release_date'],
        format='%m/%d/%y'
    )
```

```
    df.loc[df['release_date'].dt.year > 2019, 'release_date'] -= pd

    return df

train = parse_release_dates(train)

# There is a single movie without release date in the test set, so
test.loc[3829, 'release_date'] = '03/20/01'

test = parse_release_dates(test)
```

2. EDA - Exploratory Data Analysis

2.1 Revenue

In [ ]:
```
train['log_revenue'] = np.log1p(train['revenue'])

# Plots of the revenue distribution
plt.figure(figsize=(20, 10))
plt.subplot(1, 3, 1)
sns.boxplot(x=train['revenue'])
plt.title('Boxplot of Revenue')
plt.subplot(1, 3, 2)
sns.histplot(train['revenue'], bins=50, kde=True)
plt.title('Distribution of Revenue')
plt.subplot(1, 3, 3)
sns.histplot(train['log_revenue'], bins=50, kde=True)
plt.title('Distribution of Log Revenue')
plt.tight_layout()
plt.show()
```

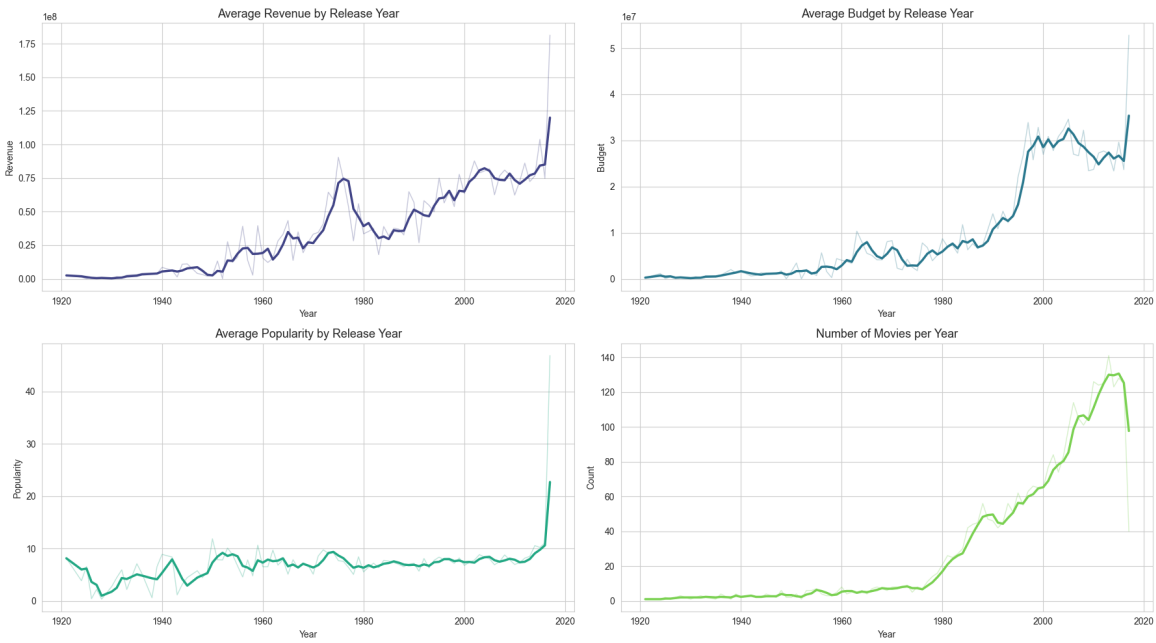


The boxplot of revenue shows that there are some outliers in the data, which is expected in a dataset like this. The histogram of revenue is right-skewed, indicating that most movies have low revenue, while a few have very high revenue. The histogram of log revenue is more normally distributed, which suggests that applying a log transformation to the target variable may help

improve the performance of our models.

2.2 Release Date

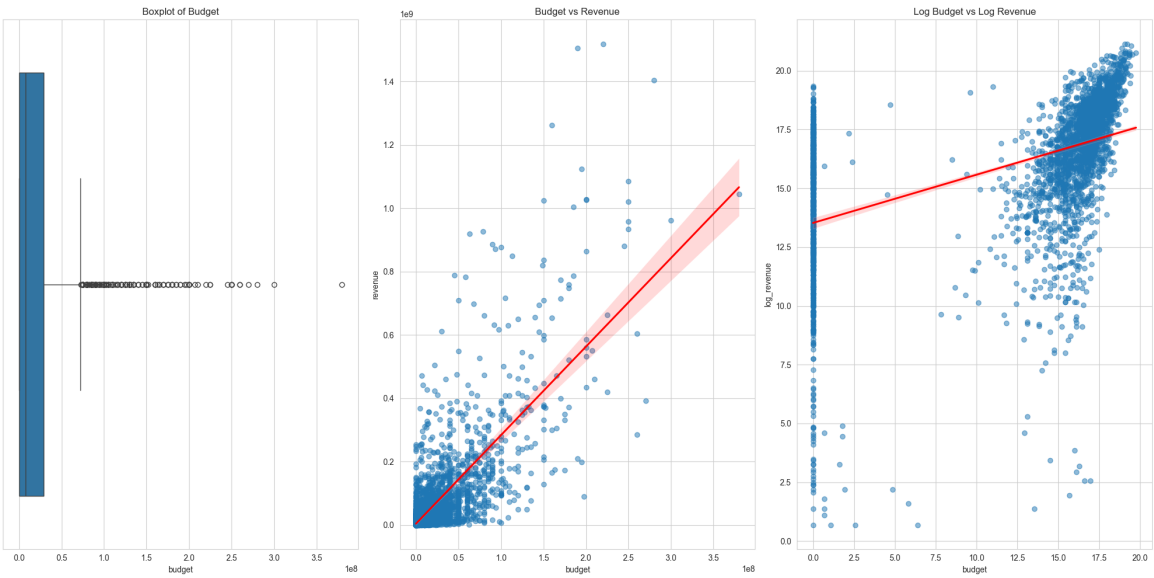
In [ ]:
```
sns.set_style("whitegrid")
release_year = pd.to_datetime(train['release_date'], errors='coerce
palette = sns.color_palette("viridis", 4)
release_year_plots = [
    {
        'data': train.groupby(release_year)['revenue'].mean().sort_
        'color': palette[0],
        'title': 'Average Revenue by Release Year',
        'ylabel': 'Revenue'
    },
    {
        'data': train.groupby(release_year)['budget'].mean().sort_i
        'color': palette[1],
        'title': 'Average Budget by Release Year',
        'ylabel': 'Budget'
    },
    {
        'data': train.groupby(release_year)['popularity'].mean().so
        'color': palette[2],
        'title': 'Average Popularity by Release Year',
        'ylabel': 'Popularity'
    },
    {
        'data': train.groupby(release_year).size().sort_index(),
        'color': palette[3],
        'title': 'Number of Movies per Year',
        'ylabel': 'Count'
    }
]

plt.figure(figsize=(18, 10))

for i, plot in enumerate(release_year_plots, 1):
    plt.subplot(2, 2, i)

    data = plot['data']
    smooth = data.rolling(window=3, min_periods=1).mean()

    sns.lineplot(x=data.index, y=data.values, color=plot['color'],
    sns.lineplot(x=smooth.index, y=smooth.values, color=plot['color

    plt.title(plot['title'], fontsize=13)
    plt.xlabel('Year')
    plt.ylabel(plot['ylabel'])

plt.tight_layout()
plt.show()
```



- **Revenue by Release Year**: Average revenue generally increases over time, with noticeable fluctuations and a sharp peak in the most recent years. This suggests that movies tend to generate higher revenues in more recent periods.
- **Budget by Release Year**: Movie budgets show a clear upward trend, especially from the 1990s onward, indicating that film production has become increasingly expensive over time.
- **Popularity by Release Year**: Popularity remains relatively stable for many decades but rises sharply in the most recent years, likely reflecting changes in how popularity is measured or the growing influence of digital platforms.
- **Movies Count per Release Year**: The number of movies released each year steadily increases, particularly after the 1980s, suggesting significant growth in film production over time. The drop in the final year is likely due to incomplete data.

2.3 Budget

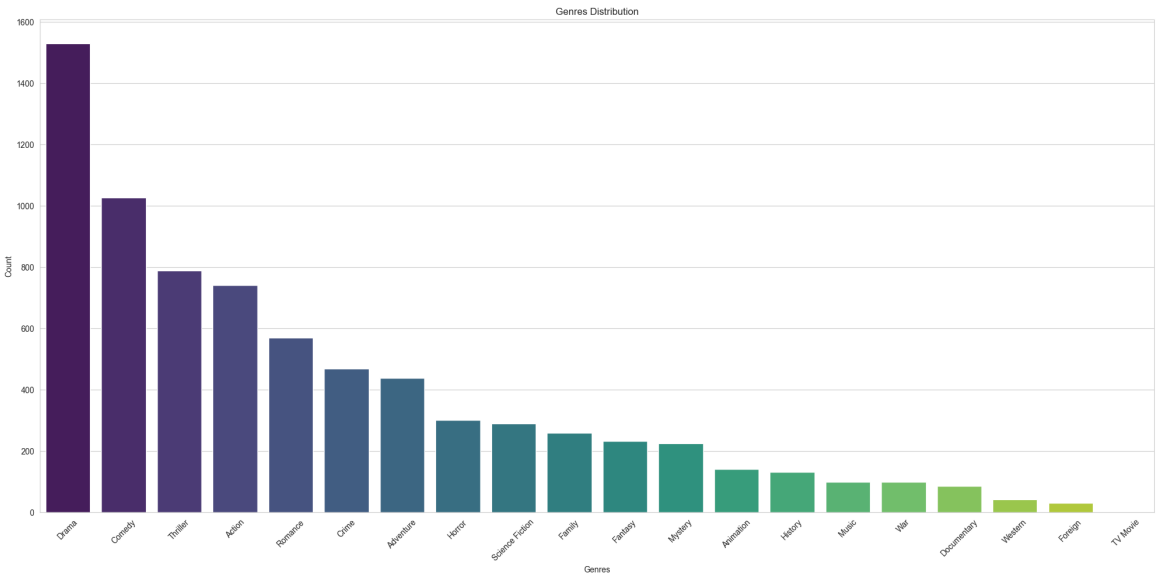
In [ ]:
```
# Plots of the Budget distribution for train set
plt.figure(figsize=(20, 10))
plt.subplot(1, 3, 1)
sns.boxplot(x=train['budget'])
plt.title('Boxplot of Budget')
plt.subplot(1, 3, 2)
sns.regplot(x='budget', y='revenue', data=train, scatter_kws={'alph
plt.title('Budget vs Revenue')
plt.subplot(1, 3, 3)
sns.regplot(x=np.log1p(train['budget']), y=train['log_revenue'], sc
plt.title('Log Budget vs Log Revenue')
plt.tight_layout()
plt.show()
```



- **Boxplot of Budget**: The budget distribution is highly right-skewed, with most movies having relatively low budgets and several extreme outliers corresponding to very expensive productions.
- **Budget vs Revenue**: There is a positive relationship between budget and revenue: higher-budget movies tend to generate higher revenues, although the variance is large, indicating that high investment does not guarantee success.
- **Log Budget vs Log Revenue**: After applying the logarithmic transformation, the relationship becomes more linear and clearer, suggesting that budget is a meaningful predictor of revenue and that the log transformation helps stabilize the variance.

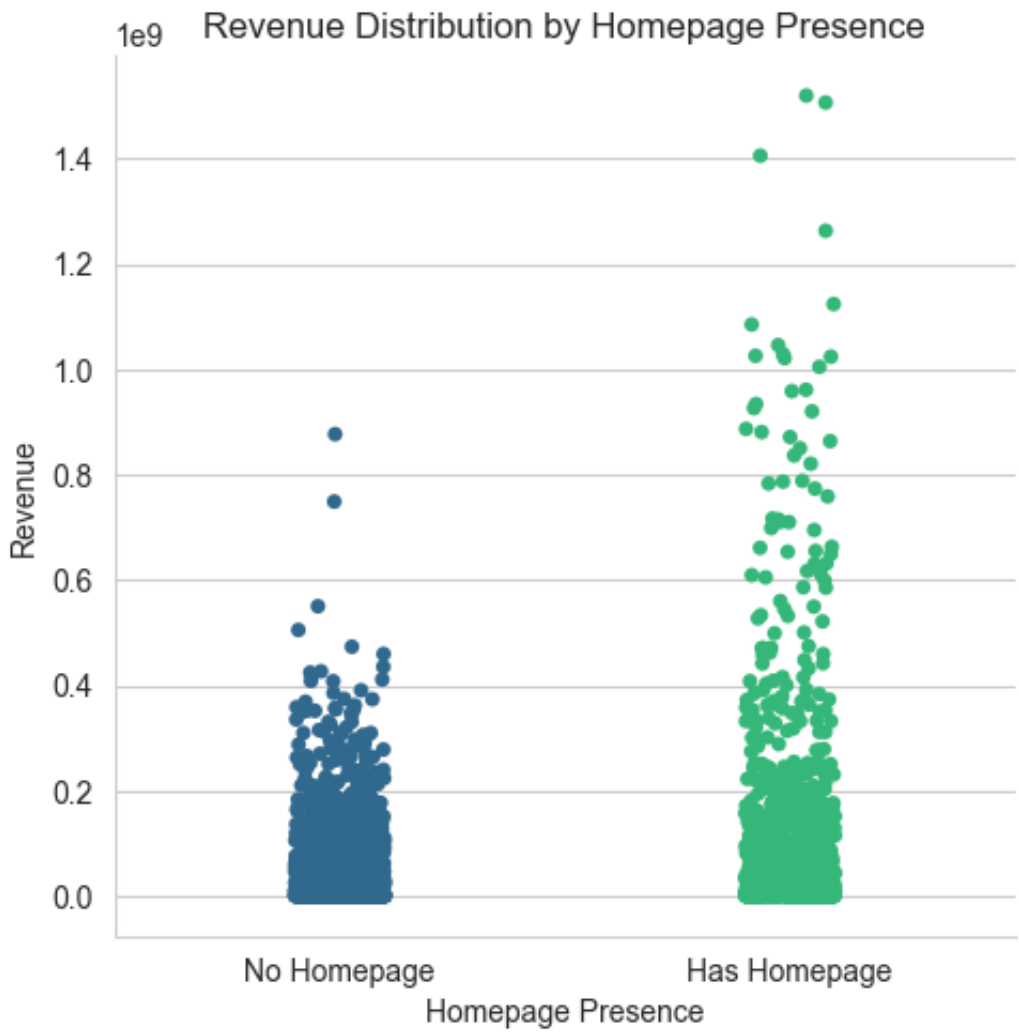
2.3 Genres

In [ ]:
```
genres = train['genres'].explode().value_counts().reset_index()
# Plots of the Genres distribution for train set
plt.figure(figsize=(20, 10))
sns.barplot(x='genres', y='count', data=genres, palette='viridis',
plt.title('Genres Distribution')
plt.xlabel('Genres')
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



2.4 Homepage

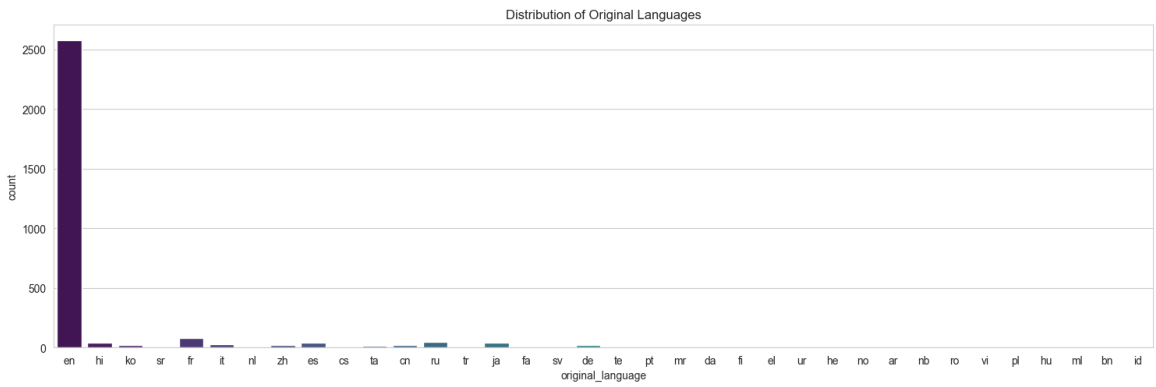
In [ ]:
```
sns.catplot(
    x=train['homepage'].notna().astype(int),
    y=train['revenue'],
    palette='viridis',
    hue=train['homepage'].notna().astype(int),
    legend=False
)
plt.title('Revenue Distribution by Homepage Presence')
plt.xticks([0, 1], ['No Homepage', 'Has Homepage'])
plt.xlabel('Homepage Presence')
plt.ylabel('Revenue')
plt.show()
```

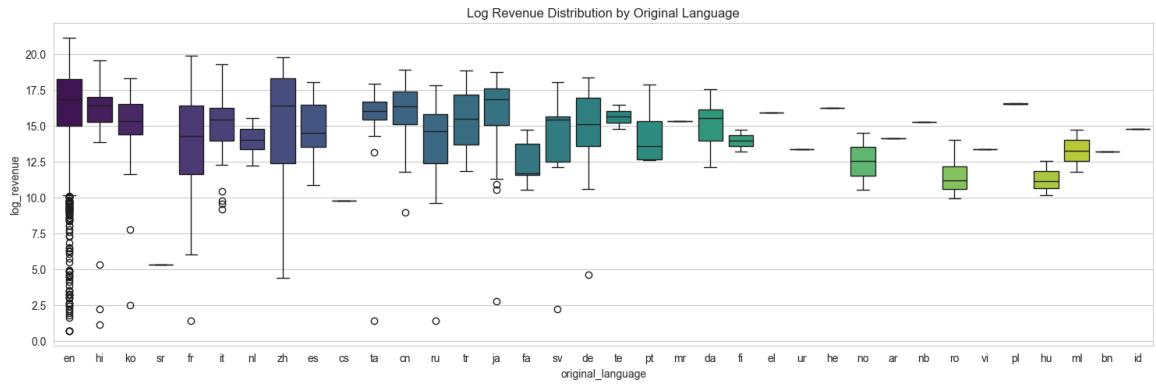


The plot shows that movies with a homepage tend to generate higher revenues compared to those without one. Films without a homepage are mostly concentrated at lower revenue levels, while movies with a homepage exhibit a wider distribution and include many of the highest-grossing titles. This suggests that the presence of a homepage may be associated with larger productions or stronger marketing efforts.

2.5 Original Language

In [ ]:
```
plt.figure(figsize=(15, 10))

plt.subplot(2, 1, 1)
plt.title("Distribution of Original Languages")
sns.countplot(x='original_language', data=train, palette='viridis',

plt.subplot(2, 1, 2)
plt.title("Log Revenue Distribution by Original Language")
sns.boxplot(x='original_language', y='log_revenue', data=train, pal

plt.tight_layout()
plt.show()
```





The plot suggests that original language may affect the revenue, with english movies showing higher medians and much greater variability. Other languages generally exhibit lower and more concentrated revenue distributions, likely reflecting more localized markets.

2.6 Production Companies

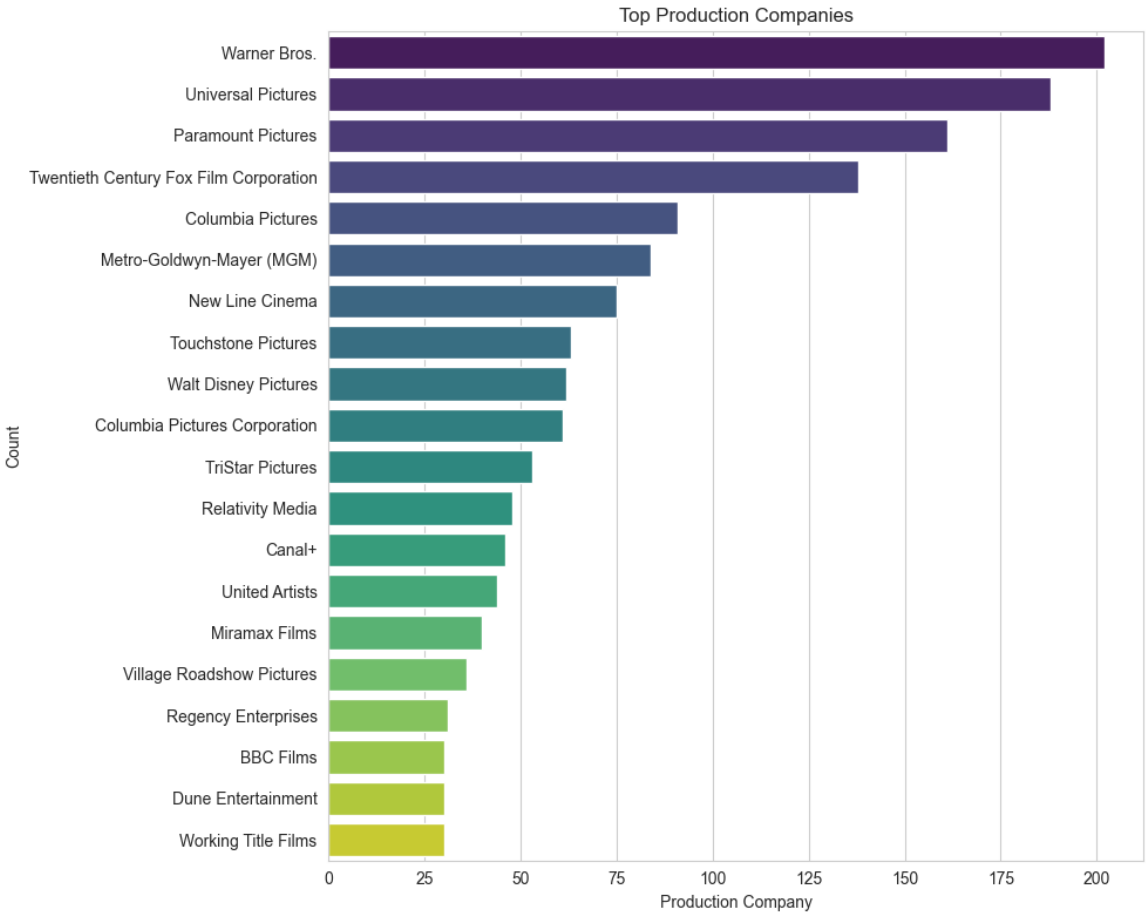
Production companies are a high-cardinality categorical feature: the dataset contains a very large number of distinct companies, many of which appear only a few times. This makes a direct one-hot encoding impractical, as it would generate a highly sparse representation and increase the risk of overfitting.

For this reason, we focus only on the most frequent production companies observed in the training set. This allows us to preserve the signal carried by major studios while controlling dimensionality. In addition, we compute the total number of production companies involved in each movie, since co-productions may provide useful information about production scale and distribution capacity.

In [ ]:
```
production_companies = train['production_companies'].explode().valu

print(f"Number of distinct production companies: {len(production_co
plt.figure(figsize = (10, 8))
sns.barplot(
    x='count',
    y='production_companies',
    data=production_companies.head(20),
    palette='viridis',
```

```
    hue='production_companies',
    legend=False
)
plt.title('Top Production Companies')
plt.xlabel('Production Company')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

```
Number of distinct production companies: 3695
```



The distribution of production companies is highly skewed: a small number of major studios appear frequently, while most companies occur only a few times. This confirms that the feature has very high cardinality and suggests that retaining only the most frequent companies is a reasonable strategy to reduce sparsity and improve generalization.

2.7 Production Countries

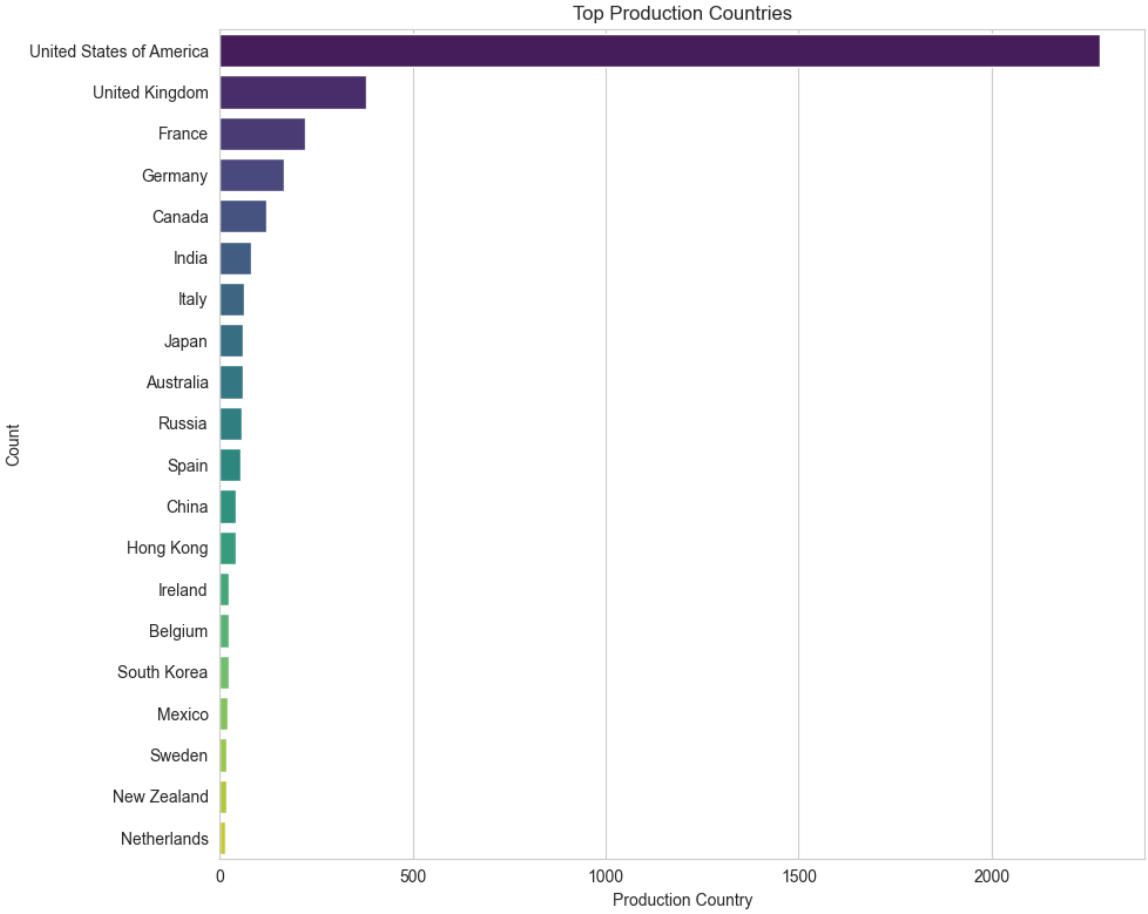
Production countries may capture differences in market size, industrial structure, and international distribution. However, this feature is also imbalanced, with a few countries appearing frequently and many others occurring only rarely.

To reduce sparsity and improve generalization, we retain only countries with a sufficient number of observations in the training data, while implicitly ignoring very rare categories. We also add the number of production countries as an

additional numerical feature, since internationally co-produced films may differ from purely domestic productions.

In [ ]:
```
production_countries = train['production_countries'].explode().valu
print(f"Number of distinct production countries: {len(production_co

plt.figure(figsize = (10, 8))
sns.barplot(
    x='count',
    y='production_countries',
    data=production_countries.head(20),
    palette='viridis',
    hue='production_countries',
    legend=False
)
plt.title('Top Production Countries')
plt.xlabel('Production Country')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```

```
Number of distinct production countries: 74
```

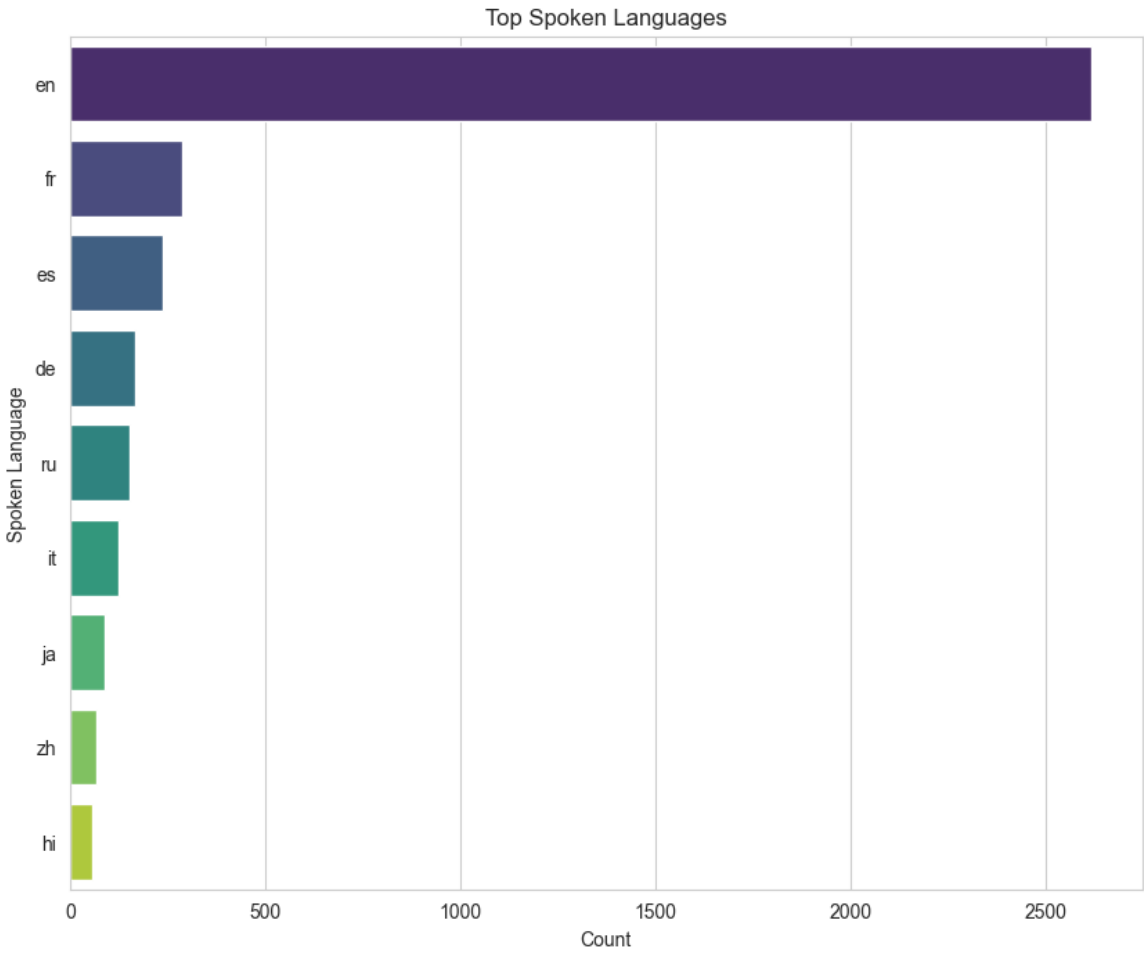


As shown in the plot, the vast majority of films were produced in the United States, while many other production countries appear only once in the dataset. To mitigate overfitting and reduce sparsity, it is reasonable to retain only countries with a sufficient number of observations (e.g., more than 30 films)

2.8 Spoken Languages

In [ ]:
```
spoken_languages = train['spoken_languages'].explode().value_counts
top_spoken_languages = spoken_languages[spoken_languages['count'] >

print(f"Number of distinct spoken languages: {len(spoken_languages)
plt.figure(figsize=(10, 8))
sns.barplot(x='count', y='spoken_languages', data=top_spoken_langua
plt.title('Top Spoken Languages')
plt.xlabel('Count')
plt.ylabel('Spoken Language')
plt.show()
```

```
Number of distinct spoken languages: 79
```

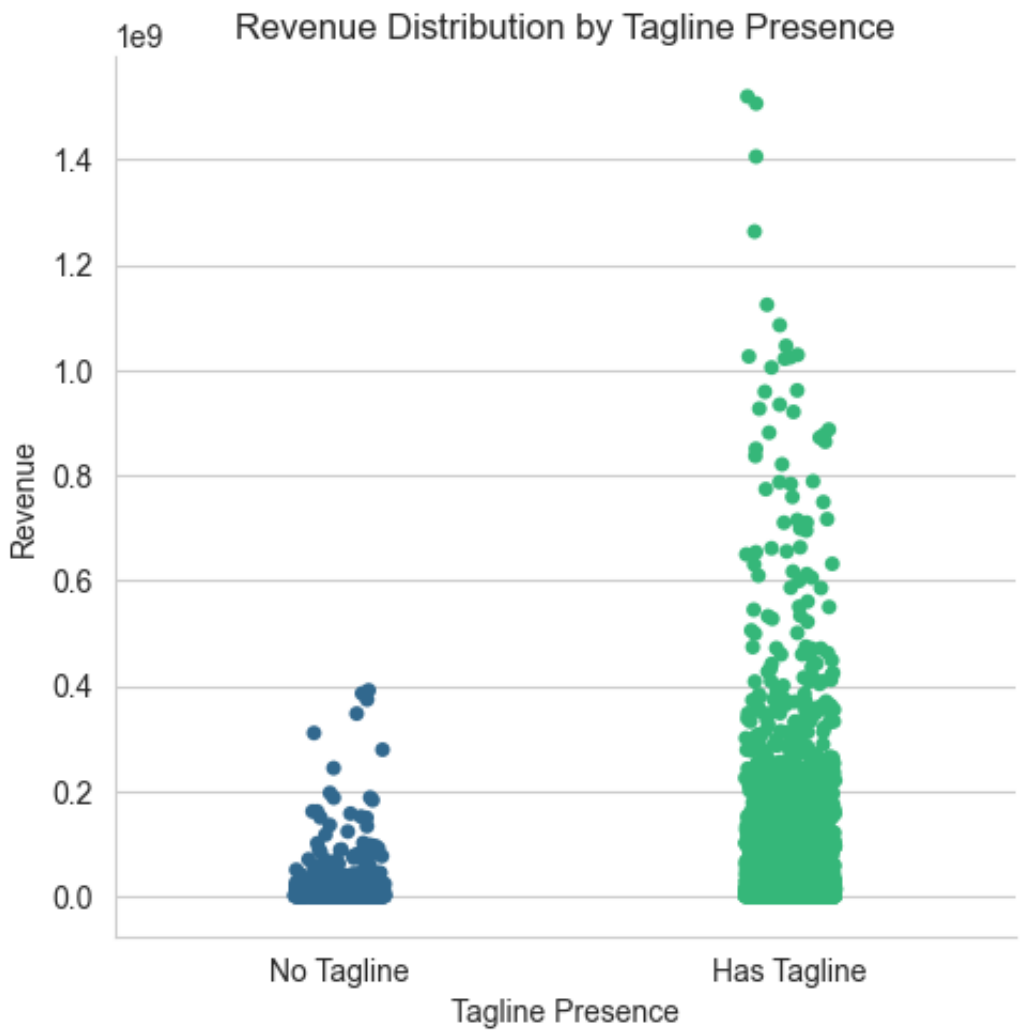


The distribution of spoken languages is highly imbalanced, with English dominating the dataset by a large margin. Other languages such as French, Spanish, German, and Russian appear much less frequently, while languages like Italian, Japanese, Chinese, and Hindi have relatively few occurrences.

2.9 Tagline

In [ ]:
```
sns.catplot(
    x=train['tagline'].notna().astype(int),
    y=train['revenue'],
    palette='viridis',
    hue=train['tagline'].notna().astype(int),
```

```
    legend=False
)
plt.title('Revenue Distribution by Tagline Presence')
plt.xticks([0, 1], ['No Tagline', 'Has Tagline'])
plt.xlabel('Tagline Presence')
plt.ylabel('Revenue')
plt.show()
```



The plot shows that movies with a tagline tend to achieve higher revenues compared to those without one. Films without a tagline are mostly concentrated at lower revenue values, while movies with a tagline display a wider spread and include many high-revenue outliers. This suggests that the presence of a tagline may be associated with stronger promotion or higher-budget productions.

2.10 Keywords

In [ ]:
```
keywords = train['Keywords'].explode().value_counts().reset_index()

print(f"Number of distinct keywords: {len(keywords)}")

plt.figure(figsize=(15, 15))
plt.imshow(WordCloud(
```

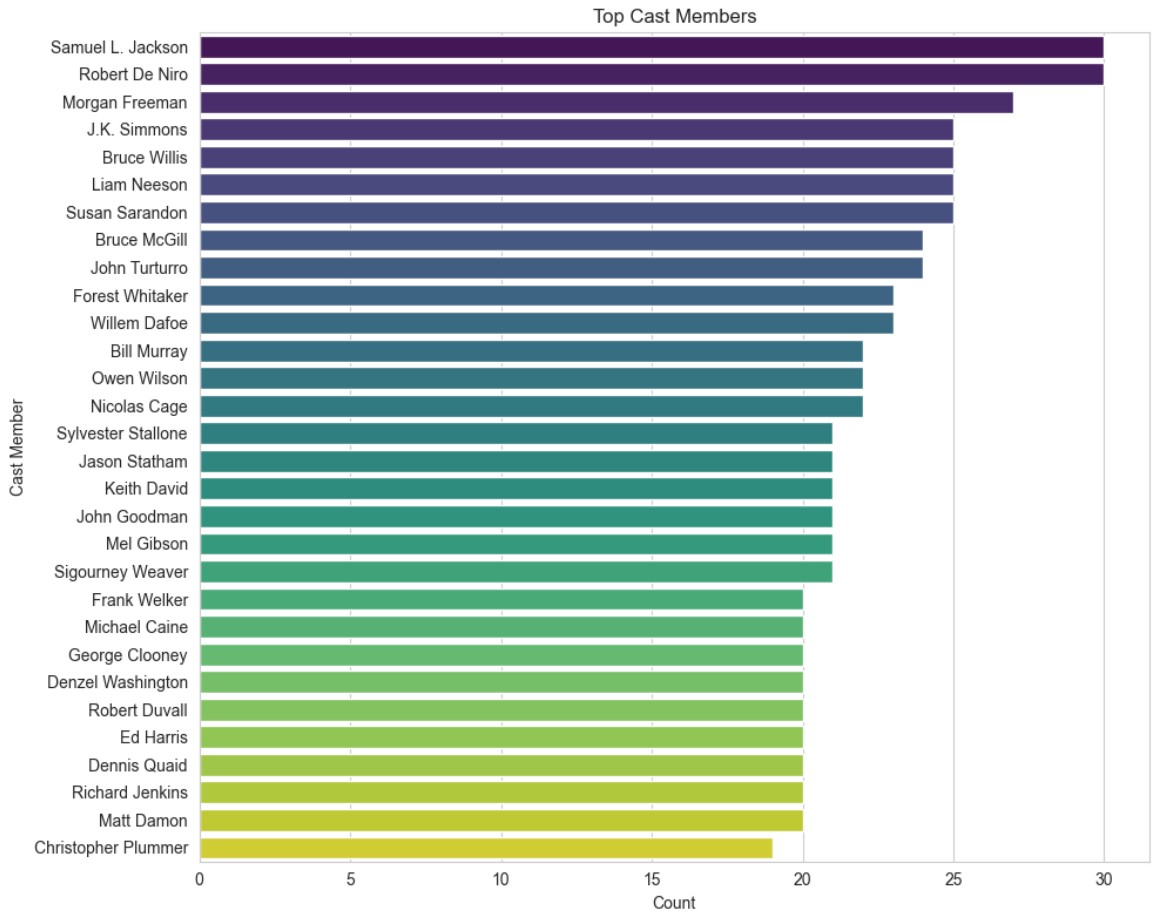
```
    width=800,
    height=400,
    background_color='white',
    colormap='viridis',
    max_words=100
).generate_from_frequencies(
    dict(zip(keywords['Keywords'], keywords['count']))
), interpolation='bilinear')
plt.axis('off')
plt.title('Word Cloud of Keywords')
plt.show()
```

```
Number of distinct keywords: 7400
```

2.11 Cast

In [ ]:
```
cast_count = train['cast'].explode().value_counts().reset_index()

print(f"Number of distinct actors: {len(cast_count)}")
plt.figure(figsize=(10, 8))
sns.barplot(x='count', y='cast', data=cast_count.head(30), palette=
plt.title('Top Cast Members')
plt.xlabel('Count')
plt.ylabel('Cast Member')
plt.tight_layout()
plt.show()
```

```
Number of distinct actors: 38588
```



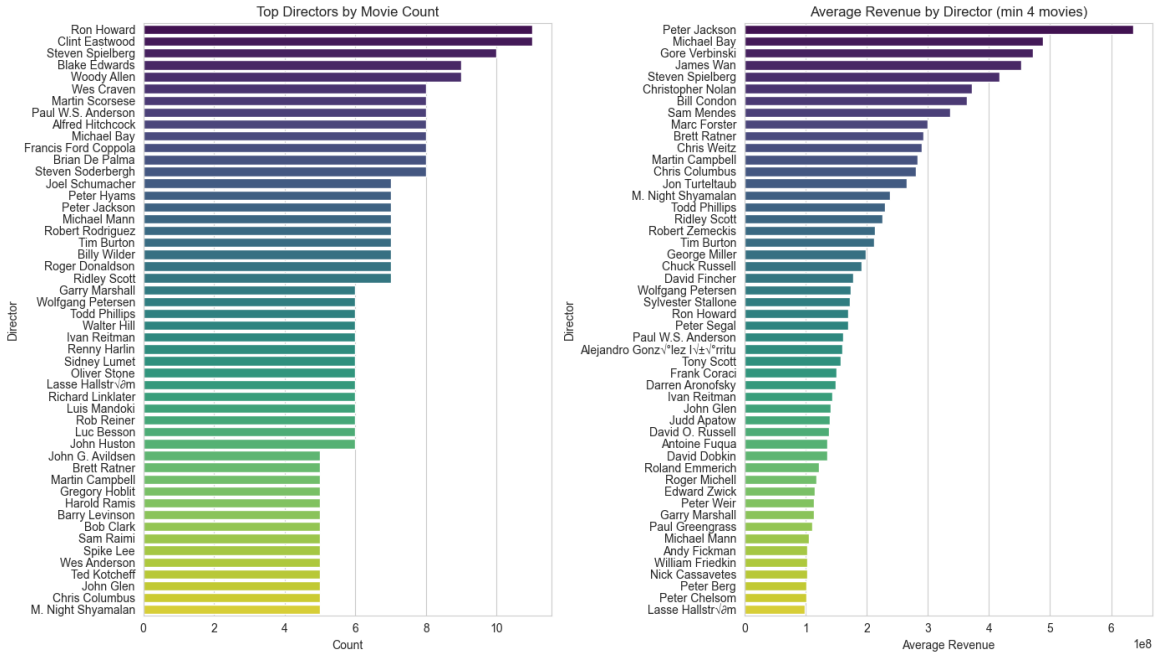
2.12 Crew

In [ ]:

```
directors_count = train['director'].value_counts()
top_directors = directors_count[directors_count >= 4]

print(f"Number of distinct directors: {train['director'].nunique()}

plt.figure(figsize=(14,8))
plt.subplot(1,2,1)
sns.barplot(
    x=top_directors.head(50).values,
    y=top_directors.head(50).index,
    palette="viridis",
    hue=top_directors.head(50).index,
    legend=False
)

plt.title("Top Directors by Movie Count")
plt.xlabel("Count")
plt.ylabel("Director")

director_revenue = (
    train[train['director'].isin(top_directors.index)]
    .groupby('director')['revenue']
    .mean()
    .sort_values(ascending=False)
    .head(50)
)
```

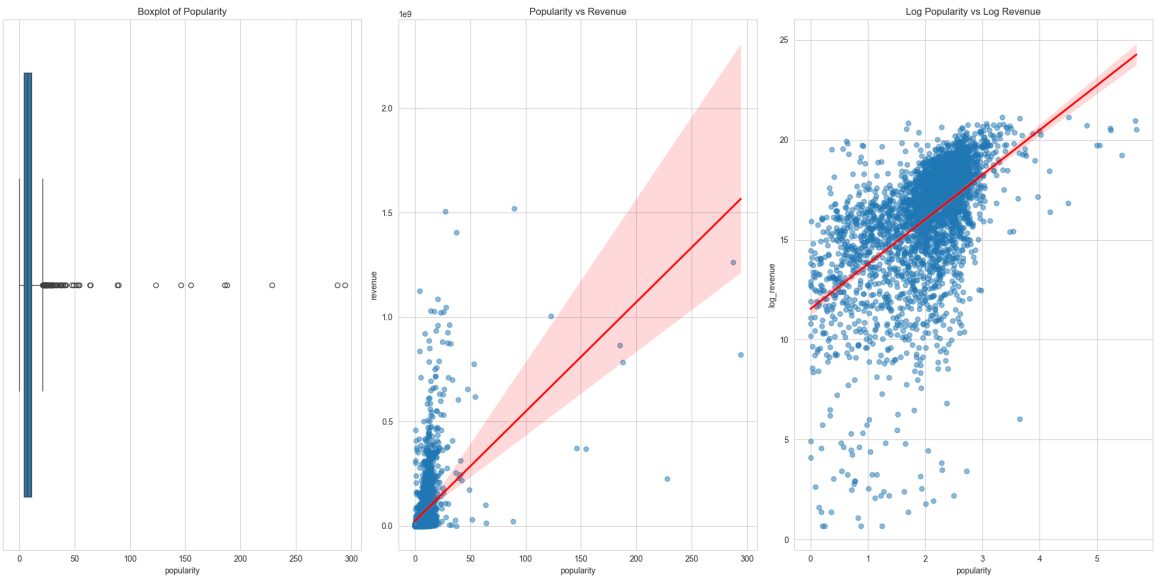
```
plt.subplot(1,2,2)

sns.barplot(
    x=director_revenue.values,
    y=director_revenue.index,
    palette="viridis",
    hue=director_revenue.index,
    legend=False
)

plt.title("Average Revenue by Director (min 4 movies)")
plt.xlabel("Average Revenue")
plt.ylabel("Director")

plt.tight_layout()
plt.show()
```

```
Number of distinct directors: 1857
```



2.13 Popularity

In [ ]:
```
# Plots of the Popularity distribution for train set
plt.figure(figsize=(20, 10))
plt.subplot(1, 3, 1)
sns.boxplot(x=train['popularity'])
plt.title('Boxplot of Popularity')
plt.subplot(1, 3, 2)
sns.regplot(x='popularity', y='revenue', data=train, scatter_kws={'
plt.title('Popularity vs Revenue')
plt.subplot(1, 3, 3)
sns.regplot(x=np.log1p(train['popularity']), y=train['log_revenue']
plt.title('Log Popularity vs Log Revenue')
plt.tight_layout()
plt.show()
```



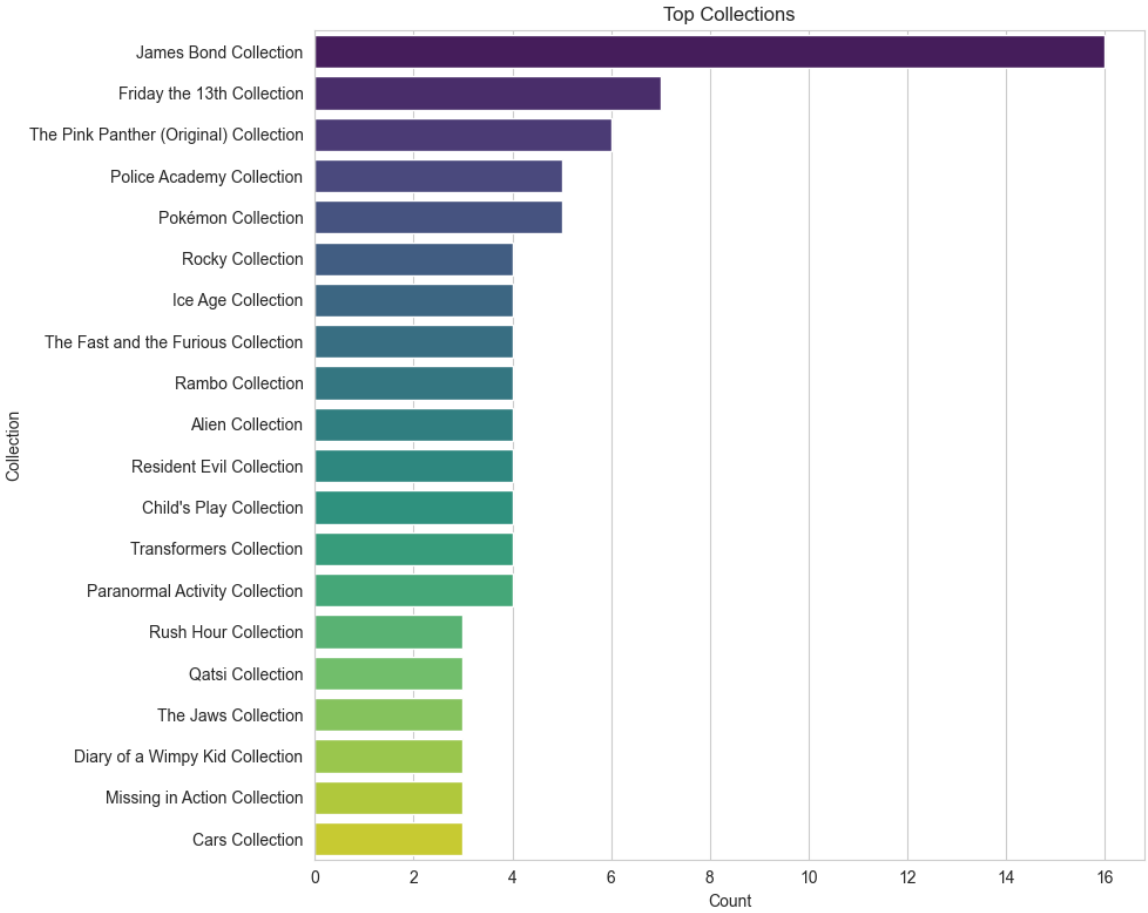
- **Boxplot of Popularity**: The popularity distribution is strongly right-skewed, with most movies having relatively low popularity and several extreme outliers with very high values.
- **Popularity vs Revenue**: There is a positive relationship between popularity and revenue, although the dispersion is large, indicating that highly popular movies tend to earn more but with considerable variability.
- **Log Popularity vs Log Revenue**: After applying the logarithmic transformation, the relationship becomes more linear and clearer, suggesting that popularity is a useful predictor of revenue and that the log transformation helps stabilize the variance.

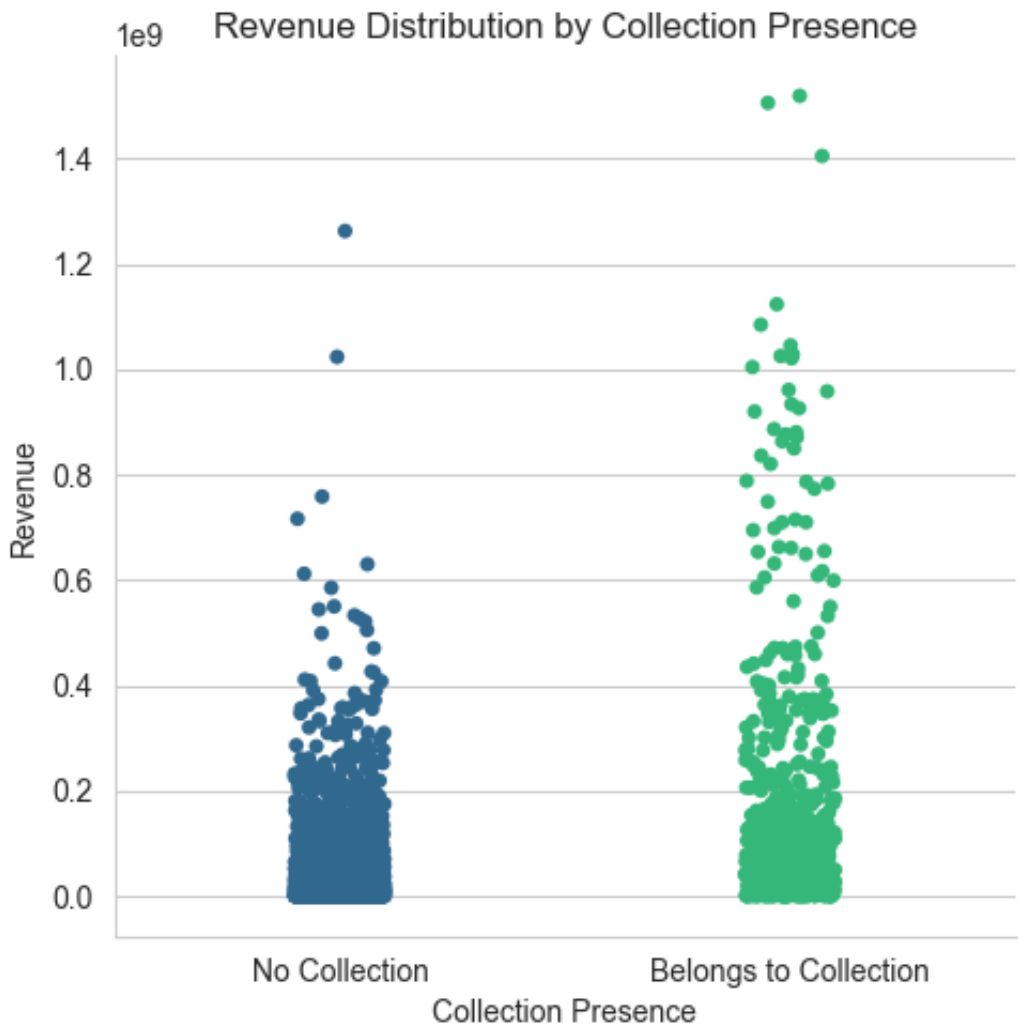
2.14 Belongs To Collection

In [ ]:
```
collections = train['belongs_to_collection'].explode().value_counts

print(f"Number of distinct collections: {len(collections)}")
plt.figure(figsize=(10, 8))
sns.barplot(x='count', y='belongs_to_collection', data=collections.
plt.title('Top Collections')
plt.xlabel('Count')
plt.ylabel('Collection')
plt.tight_layout()
plt.show()

collections_analysis = train.copy()
collections_analysis['belongs_to_collection'] = collections_analysi

sns.catplot(
    x=collections_analysis['belongs_to_collection'],
    y=collections_analysis['revenue'],
    palette='viridis',
    hue=collections_analysis['belongs_to_collection'],
    legend=False
)
```

```
plt.title('Revenue Distribution by Collection Presence')
plt.xticks([0, 1], ['No Collection', 'Belongs to Collection'])
plt.xlabel('Collection Presence')
plt.ylabel('Revenue')
plt.show()
```

```
Number of distinct collections: 422
```





The plot shows that movies belonging to a collection tend to generate higher revenues compared to standalone films. While movies without a collection are mostly concentrated at lower revenue levels, films that are part of a collection exhibit a wider distribution and include many of the highest-grossing titles. This suggests that franchises or sequels are often associated with greater commercial success.

3. Feature Engineering

In [ ]:
```
X = train.drop(columns=['revenue', 'log_revenue'])
y = train['log_revenue']

X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=t
print(f"Size of the training set: {X_train.shape[0]} samples")
print(f"Size of the validation set: {X_val.shape[0]} samples")
```

```
Size of the training set: 2400 samples
Size of the validation set: 600 samples
```

3.1 Release Date

The revenue of a movie may be influenced by its release date, particularly

when it is released during specific periods of the year associated with higher audience demand. For this reason, we extract a set of informative temporal features from the release date to capture potential seasonal effects and release timing patterns.

In [ ]:
```
def process_release_date(df):
    df = df.copy()

    df['release_year'] = df['release_date'].dt.year
    df['release_month'] = df['release_date'].dt.month
    df['release_day'] = df['release_date'].dt.day
    df['release_weekday'] = df['release_date'].dt.weekday

    month = df['release_month']
    df['is_summer_release'] = month.isin([5, 6, 7, 8]).astype(int)
    df['is_holiday_release'] = month.isin([11, 12]).astype(int)
    df['is_awards_season_release'] = month.isin([10, 11, 12]).astyp
    df['is_dump_month_release'] = month.isin([1, 2, 9]).astype(int)

    df['release_quarter'] = ((month - 1) // 3 + 1).astype(int)

    df = df.drop(columns=['release_date'])

    return df

X_train = process_release_date(X_train)
X_val = process_release_date(X_val)
test = process_release_date(test)
```

3.2 Belongs To Collection

In [ ]:
```
def process_belongs_to_collection(df):
    df['belongs_to_collection'] = df['belongs_to_collection'].apply
    return df
```

3.3 Budget

Since a large number of movies in the dataset have a budget equal to zero, which is unrealistic for actual film productions, these values are likely placeholders for missing data rather than true budgets. To address this issue, we replace zero budgets with the **median budget of movies released in the same year**.

Using the median instead of the mean makes the imputation more robust to extreme values, since movie budgets are highly skewed: a small number of blockbuster films can have budgets that are orders of magnitude larger than typical productions. Additionally, imputing the budget based on the **release year** accounts for changes in production costs over time (e.g., inflation and industry growth), ensuring that the imputed value is consistent with the

economic context of that period.

In [ ]:
```
median_budget_per_year = X_train.loc[X_train['budget'] > 0].groupby
global_budget_median = X_train.loc[X_train['budget'] > 0, 'budget']

def budget_preprocessing(df):
    df['budget'] = df['budget'].astype(float)
    df['budget_is_zero'] = (df['budget'] == 0).astype(int)

    year_median = df['release_year'].map(median_budget_per_year)
    year_median = year_median.fillna(global_budget_median)

    df.loc[df['budget'] == 0, 'budget'] = year_median[df['budget']
    df['log_budget'] = np.log1p(df['budget'])

    return df
```

3.4 Genres

Genres are naturally a multi-label categorical feature, since a movie may belong to more than one genre at the same time. Therefore, a standard one-hot encoding would not be appropriate in its usual single-category form.

To represent this information, we use a multi-hot encoding through `MultiLabelBinarizer`, creating one binary feature for each genre. This preserves the full genre composition of each movie and allows the models to learn which genres, or combinations of genres, are associated with higher or lower revenues.

In [ ]:
```
def process_genres(df, mlb):
    genre_encoded = mlb.transform(df['genres'])
    genre_df = pd.DataFrame(genre_encoded, columns=mlb.classes_, in
    df = pd.concat([df.drop(columns=['genres']), genre_df], axis=1)
    return df

genres_mlb = MultiLabelBinarizer()
genres_mlb.fit(X_train['genres'])
```

Out[ ]:
```
▾ MultiLabelBinarizer
▸ Parameters
```

3.5 Homepage

In [ ]:
```
def process_homepage(df):
    df['has_homepage'] = df['homepage'].notna().astype(int)
    df = df.drop(columns=['homepage'])
    return df
```

3.6 Original Language

In [ ]:
```
def map_original_languages(df, top_original_languages):
    df['original_language'] = df['original_language'].where(
        df['original_language'].isin(top_original_languages),
        'other'
    )
    return df

def encode_original_languages(df, encoder):
    encoded_cols = encoder.get_feature_names_out(['original_languag
    encoded_df = pd.DataFrame(encoder.transform(df[['original_langu
    df = pd.concat([df.drop(columns=['original_language']), encoded
    return df

top_original_languages = X_train['original_language'].value_counts(
print(f"Top 10 Original Languages:\n{top_original_languages}")

X_train = map_original_languages(X_train, top_original_languages)
original_language_ohe = OneHotEncoder(sparse_output=False, handle_u
original_language_ohe.fit(X_train[['original_language']])
```

```
Top 10 Original Languages:
['en', 'fr', 'ru', 'hi', 'es', 'ja', 'it', 'ko', 'de', 'cn']
```

Out[ ]:

```
▾ OneHotEncoder
▸ Parameters
```

3.7 Production Companies

The `production_companies` field contains a large number of distinct labels, making it a high-cardinality multi-label feature. Encoding every company would lead to an excessively sparse design matrix and would likely introduce noise from very infrequent categories.

To address this issue, we keep only the top 20 most frequent production companies in the training set and encode them using `MultiLabelBinarizer`. This preserves the information associated with the most relevant studios while limiting dimensionality. We also create an additional feature representing the total number of production companies involved in each film.

In [ ]:
```
def transform_production_companies(df, mlb):
    df['num_production_companies'] = df['production_companies'].app

    filtered_companies = df['production_companies'].apply(
        lambda x: [k for k in x if k in mlb.classes_] if isinstance
    )

    encoded = mlb.transform(filtered_companies)
```

```
    production_companies_df = pd.DataFrame(
        encoded,
        columns=mlb.classes_,
        index=df.index
    ).add_prefix('production_company_')

    return pd.concat([df.drop(columns=['production_companies']), pr

production_companies = X_train['production_companies'].explode().va
top_companies = production_companies.head(20).index.tolist()

production_companies_mlb = MultiLabelBinarizer(classes=top_companie
production_companies_mlb.fit(X_train['production_companies'])

print(f"The top {len(top_companies)} production companies are {top_
```

```
The top 20 production companies are ['Warner Bros.', 'Universal Pict
ures', 'Paramount Pictures', 'Twentieth Century Fox Film Corporatio
n', 'Columbia Pictures', 'Metro-Goldwyn-Mayer (MGM)', 'New Line Cine
ma', 'Touchstone Pictures', 'Columbia Pictures Corporation', 'Walt D
isney Pictures', 'TriStar Pictures', 'Canal+', 'Relativity Media', '
United Artists', 'Village Roadshow Pictures', 'Miramax Films', 'Rege
ncy Enterprises', 'Dune Entertainment', 'Fox Searchlight Pictures',
'Lionsgate']
```

3.8 Production Countries

Production countries are encoded as a multi-label categorical feature because a movie can be associated with more than one country. As with production companies, the distribution is highly imbalanced, with a small number of countries accounting for most observations.

To reduce sparsity and avoid overfitting, we retain only countries appearing more than 30 times in the training set. The selected countries are then encoded with `MultiLabelBinarizer`. In addition, we compute the number of production countries for each movie as a separate numerical feature.

In [ ]:
```
def transform_production_countries(df, mlb):
    df['num_production_countries'] = df['production_countries'].app

    filtered_countries = df['production_countries'].apply(
        lambda x: [k for k in x if k in mlb.classes_] if isinstance
    )

    encoded = mlb.transform(filtered_countries)
    production_countries_df = pd.DataFrame(
        encoded,
        columns=mlb.classes_,
        index=df.index
    ).add_prefix('production_country_')

    return pd.concat([df.drop(columns=['production_countries']), pr
```

```
production_countries = X_train['production_countries'].explode().va
top_countries = production_countries[production_countries > 30].ind

production_countries_mlb = MultiLabelBinarizer(classes=top_countrie
production_countries_mlb.fit(X_train['production_countries'])

print(f"The top {len(top_countries)} production countries are {top_
```

```
The top 13 production countries are ['United States of America', 'Un
ited Kingdom', 'France', 'Germany', 'Canada', 'India', 'Australia',
'Italy', 'Russia', 'Japan', 'Spain', 'China', 'Hong Kong']
```

3.9 Spoken Languages

In [ ]:
```
def transform_spoken_languages(df, mlb):
    df['num_spoken_languages'] = df['spoken_languages'].apply(len)

    filtered_spoken_languages = df['spoken_languages'].apply(
        lambda x: [k for k in x if k in mlb.classes_] if isinstance
    )

    encoded = mlb.transform(filtered_spoken_languages)
    spoken_languages_df = pd.DataFrame(
        encoded,
        columns=mlb.classes_,
        index=df.index
    ).add_prefix('spoken_language_')

    return pd.concat([df.drop(columns=['spoken_languages']), spoken

spoken_languages = X_train['spoken_languages'].explode().value_coun
top_spoken_languages = spoken_languages[spoken_languages > 30].inde

spoken_languages_mlb = MultiLabelBinarizer(classes=top_spoken_langu
spoken_languages_mlb.fit(X_train['spoken_languages'])

print(f"The top {len(top_spoken_languages)} spoken languages are {t
```

```
The top 12 spoken languages are ['en', 'fr', 'es', 'de', 'ru', 'it',
'ja', 'zh', 'hi', 'ar', 'pt', 'ko']
```

3.10 Tagline

In [ ]:
```
def process_tagline(df):
    df['has_tagline'] = df['tagline'].apply(lambda x: 0 if pd.isna(
    return df.drop(columns=['tagline'])
```

3.11 Keywords

Keywords summarize themes, topics, and semantic elements of a movie, and may therefore provide valuable predictive information about box office performance. However, this field has extremely high cardinality, with

thousands of distinct keywords in the dataset.

To prevent the feature space from becoming excessively sparse, we retain only the most frequent keywords and encode them with `MultiLabelBinarizer`. We also compute the total number of keywords assigned to each movie, which may reflect the richness or specificity of its metadata representation.

In [ ]:
```
def process_keywords(df, mlb):
    df['num_keywords'] = df['Keywords'].apply(
        lambda x: len(x) if isinstance(x, list) else 0
    )

    filtered_keywords = df['Keywords'].apply(
        lambda x: [k for k in x if k in mlb.classes_] if isinstance
    )

    encoded = mlb.transform(filtered_keywords)

    keywords_df = (
        pd.DataFrame(encoded, columns=mlb.classes_, index=df.index)
        .add_prefix("keyword_")
    )

    df = pd.concat([df, keywords_df], axis=1)

    return df.drop(columns=['Keywords'])

keywords_counts = X_train['Keywords'].explode().value_counts()
top_keywords = keywords_counts[keywords_counts > 50].index.tolist()

keywords_mlb = MultiLabelBinarizer(classes=top_keywords)
keywords_mlb.fit(X_train['Keywords'])

print(f"The top {len(top_keywords)} keywords are:\n{top_keywords}")
```

```
The top 15 keywords are:
['woman director', 'independent film', 'duringcreditsstinger', 'murd
er', 'based on novel', 'sport', 'violence', 'revenge', 'dystopia', '
friendship', 'aftercreditsstinger', 'sex', 'biography', 'love', 'sus
pense']
```

3.12 Cast

The cast information may capture the presence of well-known actors, which can influence audience interest and commercial performance. At the same time, the number of distinct cast members is extremely large, making a full encoding infeasible.

For this reason, we retain only the most frequent cast members in the training set and encode them as binary indicators using `MultiLabelBinarizer`. We additionally compute the number of cast members listed for each movie,

since ensemble size may also contain useful information.

In [ ]:
```
cast_count = X_train['cast'].explode().value_counts()
top_cast = cast_count[cast_count > 15].index.tolist()

def process_cast(df, mlb):
    filtered_cast = df['cast'].apply(
        lambda x: [k for k in x if k in mlb.classes_] if isinstance
    )

    encoded = mlb.transform(filtered_cast)

    cast_df = pd.DataFrame(
        encoded,
        columns=mlb.classes_,
        index=df.index
    ).add_prefix('cast_')

    df = pd.concat([df.drop(columns=['cast']), cast_df], axis=1)
    return df

cast_mlb = MultiLabelBinarizer(classes=top_cast)
cast_mlb.fit(X_train['cast'])

print(f"The top {len(top_cast)} cast members are:\n{top_cast}")
```

```
The top 29 cast members are:
['Samuel L. Jackson', 'Morgan Freeman', 'Liam Neeson', 'J.K. Simmon
s', 'Robert De Niro', 'Bruce Willis', 'Bill Murray', 'Forest Whitake
r', 'Willem Dafoe', 'Jason Statham', 'John Turturro', 'Bruce McGil
l', 'Susan Sarandon', 'Ed Harris', 'Robert Duvall', 'Sylvester Stall
one', 'Denzel Washington', 'Philip Seymour Hoffman', 'Kevin Bacon',
'Steve Buscemi', 'Christopher Plummer', 'Nicolas Cage', 'Allison Jan
ney', 'Michael Caine', 'Richard Jenkins', 'John Goodman', 'George Cl
ooney', 'Christopher Walken', 'Dennis Quaid']
```

3.13 Crew

The `crew` field contains many different professional roles, but not all of them are equally informative for revenue prediction. Among them, the director is one of the most interpretable and potentially relevant signals, as directors may capture style, reputation, and production quality.

We therefore extract the director from the crew list and encode only the most frequent directors, grouping the remaining ones into an `other` category. This reduces the cardinality of the feature while preserving the information associated with directors who appear sufficiently often in the training set.

In [ ]:
```
def map_director(df, top_directors):
    df = df.copy()
    df['director'] = df['director'].apply(lambda x: x if x in top_d
    return df
```

```
def process_crew(df, one_hot_encoder):
    df = df.copy()
    # crew size feature
    df['num_crew'] = df['crew'].apply(lambda x: len(x) if isinstanc
    # encode director
    director_encoded = one_hot_encoder.transform(df[['director']])
    director_df = pd.DataFrame(
        director_encoded,
        columns=one_hot_encoder.get_feature_names_out(['director'])
        index=df.index
    ).astype(int)

    df = pd.concat([df.drop(columns=['director', 'crew']), director

    return df

# Top Directors
directors_count = X_train['director'].value_counts()
top_directors = directors_count[directors_count > 5].index.tolist()

X_train = map_director(X_train, top_directors)

director_ohe = OneHotEncoder(
    sparse_output=False,
    handle_unknown='ignore'
)
director_ohe.fit(X_train[['director']])
```

Out[ ]:

▾ OneHotEncoder ⓘ ?

▸ Parameters

3.14 Popularity

In [ ]:

```
def process_popularity(df):
    df['log_popularity'] = np.log1p(df['popularity'])
    return df
```

3.15 Runtime

In [ ]:

```
median_runtime_per_year = X_train.loc[X_train['runtime'].notna() &
global_runtime_median = X_train.loc[X_train['runtime'].notna() & (X

def process_runtime(df):
    df = df.copy()
    df['runtime'] = df['runtime'].astype(float)

    missing_or_zero = df['runtime'].isna() | (df['runtime'] == 0)
    df['runtime_is_missing_or_zero'] = missing_or_zero.astype(int)

    year_median = df['release_year'].map(median_runtime_per_year).f
    df.loc[missing_or_zero, 'runtime'] = year_median[missing_or_zer
```

```
    df['log_runtime'] = np.log1p(df['runtime'])
    return df
```

3.16 Cross Features

In addition to the original predictors, we construct several interaction and ratio-based features in order to capture non-linear relationships between budget, popularity, and runtime. These variables aim to provide the models with more expressive information than the raw features alone.

For example, the interaction between log-budget and log-popularity may capture the combined effect of financial investment and public attention. Ratios such as budget per runtime or popularity per runtime normalize the original quantities and may help distinguish short, highly promoted films from longer productions with similar absolute values. These engineered features are intended to improve predictive performance, especially for tree-based models.

In [ ]:
```
def cross_features(df):
    df['log_budget_x_log_popularity'] = df['log_budget'] * df['log_
    df['budget_per_runtime'] = df['budget'] / df['runtime']
    df['popularity_per_runtime'] = df['popularity'] / df['runtime']
    df['popularity_to_budget'] = df['popularity'] / (df['budget'] +
    df['log_budget_per_runtime'] = df['log_budget'] / np.log1p(df['

    return df
```

3.17 Other Features

In [ ]:
```
def process_other_features(df):
    df.columns = df.columns.str.lower().str.strip().str.replace(' '
    return df.drop(columns=['imdb_id', 'title', 'original_title', '
```

3.18 Preprocessing Pipeline

In [ ]:
```
def preprocessing_pipeline(df):
    df = df.copy()
    # Belongs to collection
    df = process_belongs_to_collection(df)
    # Budget
    df = budget_preprocessing(df)
    # Genres
    df = process_genres(df, genres_mlb)
    # Homepage
    df = process_homepage(df)
    # Original Language
    df['original_language'] = df['original_language'].where(df['ori
    df = encode_original_languages(df, original_language_ohe)
```

```
    # Production Companies
    df = transform_production_companies(df, production_companies_ml
    # Production Countries
    df = transform_production_countries(df, production_countries_ml
    # Spoken Languages
    df = transform_spoken_languages(df, spoken_languages_mlb)
    # Tagline
    df = process_tagline(df)
    # Keywords
    df = process_keywords(df, keywords_mlb)
    # Cast
    df = process_cast(df, cast_mlb)
    # Crew
    df = map_director(df, top_directors)
    df = process_crew(df, director_ohe)
    # Popularity
    df = process_popularity(df)
    # Runtime
    df = process_runtime(df)
    # Cross features
    df = cross_features(df)
    # Other features
    df = process_other_features(df)

    return df

X_train = preprocessing_pipeline(X_train)
X_val = preprocessing_pipeline(X_val)
X_test = preprocessing_pipeline(test)
```

4. Model Tuning

In this section, we compare three tree-based regression models: Random Forest, XGBoost, and CatBoost. These algorithms are well suited to tabular data and can naturally model non-linear relationships and complex feature interactions, which are expected in a box office prediction task.

Hyperparameter tuning is performed using `RandomizedSearchCV` with 5-fold cross-validation. This approach allows us to explore a broad hyperparameter space while keeping the computational cost manageable. Since the target variable has been log-transformed, model selection is carried out on the transformed scale, which is consistent with the objective of reducing the impact of extreme revenue values.

In [ ]:
```
results = []

from sklearn.metrics import mean_squared_log_error, make_scorer

def rmsle(y_true, y_pred):
    return np.sqrt(mean_squared_log_error(np.expm1(y_true), np.expm

rmsle_scorer = make_scorer(
```

```
    lambda y_true, y_pred: rmsle(y_true, y_pred),
    greater_is_better=False
)
cv = KFold(n_splits=5, shuffle=True, random_state=42)
```

4.1 Random Forest Regressor

Random Forest is used as a strong non-parametric baseline. Its ensemble structure makes it robust to noisy features and capable of modeling non-linear effects without requiring extensive preprocessing assumptions. The hyperparameter search focuses on tree depth, feature subsampling, sample subsampling, and minimum split constraints in order to balance bias and variance.

In [ ]:
```
rf = RandomForestRegressor(
    random_state=42,
    n_jobs=-1
)

param_dist_rf = {
    'n_estimators': [300, 500, 800],
    'max_depth': [10, 15, 20, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['sqrt', 0.6, 0.8],
    'max_samples': [0.7, 0.8, 0.9]
}

search_rf = RandomizedSearchCV(
    estimator=rf,
    param_distributions=param_dist_rf,
    n_iter=30,
    scoring=rmsle_scorer,
    cv=cv,
    verbose=1,
    random_state=42,
    n_jobs=-1
)

search_rf.fit(X_train, y_train)

rf_best = search_rf.best_estimator_
y_pred_rf = rf_best.predict(X_val)
rf_rmsle = rmsle(y_val, y_pred_rf)

results.append({
    'model': 'Random Forest Regressor',
    'rmsle': rf_rmsle,
    'r2_score': r2_score(y_val, y_pred_rf),
})

print("Best hyperparameters for Random Forest:")
print(search_rf.best_params_)
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
Best hyperparameters for Random Forest:
{'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 1,
'max_samples': 0.9, 'max_features': 0.6, 'max_depth': None}
```

4.2 XGBoost Regressor

XGBoost is a gradient boosting model designed to achieve high predictive performance on structured data. Compared with Random Forest, it typically provides greater flexibility through boosting, regularization, and shrinkage. The hyperparameter search includes tree complexity, learning rate, subsampling, column sampling, and L1/L2 regularization terms.

In [ ]:

```python
xgb_base = XGBRegressor(
    objective="reg:squarederror",
    tree_method="hist",
    random_state=42,
    n_jobs=-1
)

param_dist_xgb = {
    "n_estimators": [1500, 2000],
    "learning_rate": [0.01, 0.02, 0.03],
    "max_depth": [4, 5, 6],
    "min_child_weight": [1, 3, 5],
    "subsample": [0.6, 0.7, 0.8],
    "colsample_bytree": [0.6, 0.7, 0.8],
    "gamma": [0, 0.1, 0.3],
    "reg_alpha": [0, 0.1, 1],
    "reg_lambda": [1, 2, 3]
}

search_xgb = RandomizedSearchCV(
    estimator=xgb_base,
    param_distributions=param_dist_xgb,
    n_iter=30,
    scoring=rmsle_scorer,
    cv=cv,
    verbose=1,
    random_state=42,
    n_jobs=-1
)

search_xgb.fit(X_train, y_train)

xgb_best = XGBRegressor(
    **search_xgb.best_params_,
    objective="reg:squarederror",
    eval_metric="rmse",
    tree_method="hist",
    random_state=42,
    n_jobs=-1,
    early_stopping_rounds=100
)
```

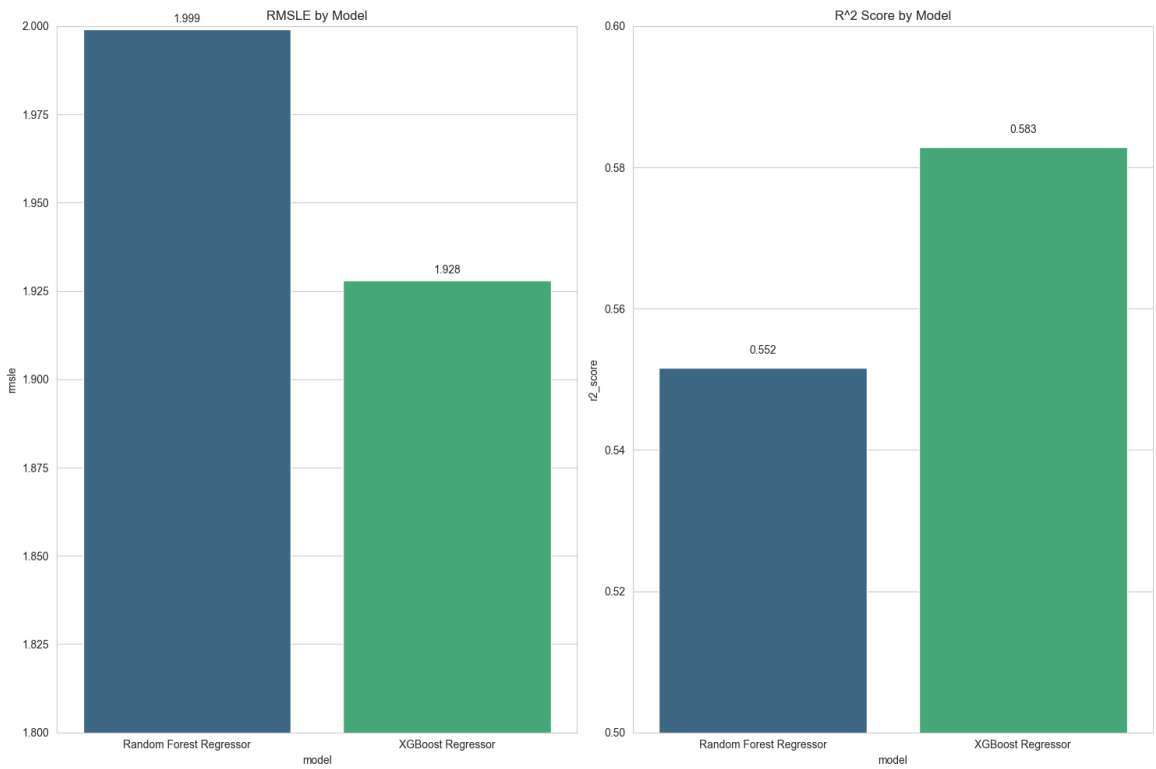
```
xgb_best.fit(
    X_train,
    y_train,
    eval_set=[(X_val, y_val)],
    verbose=False
)

y_pred_xgb = xgb_best.predict(X_val)
xgb_rmsle = rmsle(y_val, y_pred_xgb)

results.append({
    'model': 'XGBoost Regressor',
    'rmsle': xgb_rmsle,
    'r2_score': r2_score(y_val, y_pred_xgb),
})
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
```

4.5 Results

In [ ]:

```
results = pd.DataFrame(results)

plt.figure(figsize=(15, 10))

plt.subplot(1, 2, 1)
sns.barplot(x='model', y='rmsle', data=results, palette='viridis',
plt.title('RMSLE by Model')
plt.ylim(1.8, 2)
for i, v in enumerate(results["rmsle"]):
    plt.text(i, v + 0.002, f"{v:.3f}", ha='center')

plt.subplot(1, 2, 2)
sns.barplot(x='model', y='r2_score', data=results, palette='viridis
plt.title('R^2 Score by Model')
plt.ylim(.5, .6)
for i, v in enumerate(results["r2_score"]):
    plt.text(i, v + 0.002, f"{v:.3f}", ha='center')

plt.tight_layout()
plt.show()
```



5. Prediction and Submission

In [ ]:
```
y_pred = xgb_best.predict(X_test)

submission = pd.DataFrame({
    'id': X_test.index,
    'revenue': np.expm1(y_pred)
})

submission.to_csv('./data/submission.csv', index=False)
```